

[ASRJC/2024/Q1]

1 Name your Jupyter Notebook as:

TASK1_<your name>_<centre number>_<index number>.ipynb

For each of the sub-tasks, add a comment statement at the beginning of the code using the hash symbol '#', to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 1.1  
        Program code
```

Output:

A local media corporation is hosting a Top TV Artists Award. People can vote online for their favourite artist, out of 20 shortlisted ones. The voting results are stored in a file named `votes.txt`, where each line contains the voter's moniker followed by the artist they voted for, separated by a space. The following shows the first few records:

```
happypapaya Kieran_Tay  
avgstudent Alarice_Soh  
candycadet Amiyah_Kapoor
```

Task 1.1

Write program code to read the data from `votes.txt` and store the number of votes each artist received in a 2D list `results`. Each row should contain the artist's name and the corresponding vote count. [4]

Task 1.2

Write program code that applies the insertion sort to the 2D list `results`, sorting by the vote counts in **descending** order. Print `results` after sorting. [4]

Write program code to identify and output the artist(s) with the highest number of votes. Note that there can be artists sharing the same number of votes. [2]

Task 1.3

Write a function `task1_3` that uses binary search, modified to identify **all** artists that have received exactly `t` votes. Return a list containing these artists; otherwise return an empty list. Show the output for the cases where `t` is 16 and 2 respectively. [5]

Save your Jupyter Notebook for Task 1.